MIL–V–5456A
31 MARCH 1964

SUPERSEDING
MIL–P–5456
14 DECEMBER 1949

MILITARY SPECIFICATION

VACUUM PUMP, ROTARY, POWER DRIVEN

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers an engine-driven vacuum pump.

1.2 Classification. Vacuum pump shall be in accordance with the following AN part numbers, as specified (see 6.2). Pump shall have the nominal capacity and mounting provision as shown by the applicable part number:

Part No.	*Cubic feet free air per minute*	*Mounting provisions*
AN6110–1	4½	Type I pad
AN6111–1	7½	Type I pad
AN6111–2	7½	Type II pad
AN6112–1	15	Type II pad

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ–P–416 —Plating, Cadmium (Electrodeposited)
QQ–S–781 —Steel Strapping, Flat
QQ–Z–325 —Zinc Coating, Electrodeposited, Requirements for
PPP–B–585 —Boxes, Wood, Wirebound
PPP–B–601 —Boxes, Wood, Cleated-Plywood
PPP–B–621 —Boxes, Wood, Nailed and Lock-Corner
PPP–B–636 Box, Fiberboard

MILITARY

MIL–P–116 —Preservation, Methods of
MIL–C–5501 —Caps and Plugs, Protective, Dust and Moisture Seal
MIL–E–5557 Enamel; Heat-Resisting, Glyceryl-Phthalate, Black
MIL–G–5572 —Gasoline, Aviation: Grades 80/87, 91/96, 100/130, 115/145
MIL–L–6082 —Lubricating Oil; Aircraft Reciprocating Engine (Piston)
MIL–P–7105 —Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT
MIL–S–7742 —Screw Threads, Standard, Optimum Selected Series; General Specification for
MIL–P–8585 —Primer Coating, Coating, Zinc-Chromate, Low-Moisture- Sensitivity
MIL–A–8625 —Anodic-Coatings, for Aluminum and Aluminum Alloys
MIL–T–9107 —Test Reports, Preparation of

FSC 1650

Sc
Chec

MIL–C–11796 —Corrosion Preventive Compound, Petrolatum, Hot Application
MIL–C–16173 —Corrosion Preventive Compound, Solvent Cutback, Cold-Application
MIL–L–21260 —Lubricating Oil, Internal - Combustion Engine, Preservative
MIL–D–70327 —Drawings, Engineering and Associated Lists

STANDARDS

MILITARY

MIL–STD–129 —Marking for Shipment and Storage
MIL–STD–130 —Identification Marking of U. S. Military Property
MIL–STD–143 —Specifications and Standards, Order of Precedence for the Selection of

ARMY-NAVY AERONAUTICAL

AN6110 —Pump—4½ C. F. M. Engine Driven Air
AN6111 —Pump—7½ C. F. M. Engine Driven Air
AN6112 —Pump—15 C. F. M. Enine Driven Air

AIR FORCE-NAVY AERONAUTICAL

AN4044 —Gasket—Type XI Engine Accessory Drive
AN4045 —Gasket—Type X or XV Engine Accessory Drive

AIR FORCE-NAVY AERONAUTICAL DESIGN STANDARDS

AND10000 —Drive—Type I Fluid Power Pumps
AND10001 —Mounting Pad and Drive—Type II Vacuum and Hydraulic Power Pumps

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Vacuum pump design shall be as shown on the applicable AN Standard (see 1.2), as specified (see 6.2).

3.1.1 Preproduction sample. This specification requires preproduction testing. As soon as practical after award of contract or order, and prior to the submission of any complete pump units for final acceptance, two preproduction samples conforming to this specification shall be subjected to all the tests specified in 4.4.1. A preproduction sample is defined as a production line sample of the product which the manufacturer will subsequently produce in fulfillment of the contract. The preproduction samples of the vacuum pump assemblies shall be manufactured to the greatest extent possible by the use of tools, fixtures, and processes which will be used in the manufacture of additional vacuum pumps on contract or order. Approval of the preproduction samples by the procuring activity shall not relieve the contractor of his responsibility to supply equipment conforming to this specification. Any changes or deviations from the preproduction samples shall be subject to the approval of the contracting officer.

3.2 Materials. Unless otherwise specified, materials shall be selected in accordance with Standard MIL–STD–143.

3.3 Design and construction.

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3.3.1 *Direction of rotation.* The pump shall operate satisfactorily in either direction of rotation, without adjustment to the oil metering device, either external or internal. The corresponding inlet port for each direction of rotation shall be clearly and permanently marked on the pump housing.

3.3.2 *Rated speed.* The rated speed of the vacuum pump shall be 3,750 revolutions per minute (rpm).

3.3.3 *Drive.* The drive shall incorporate a flexible member having sufficient torsional flexibility to absorb torsional oscillations of the driving shaft in the engine, which may vary in magnitude up to ± 2 degrees and which may occur at frequencies above 50 cycles per second. The flexible member shall be operable in either direction of rotation. Positive stops shall prevent large deflections which would damage the flexible member. The drive shall incorporate a means for taking up misalignment to prevent overloading of the pump or drive bearings. The drive spline shall have a Rockwell C hardness of 45 to 55. If the drive spline portion of the drive is separable, it shall be retained so that a pull of 30 to 50 pounds is required to pull the drive out of the pump.

3.3.3.1 *Coupling and flange.* The vacuum pump shall be supplied with a flange and coupling for mounting directly onto the mounting pad specified in table I. Details of the spline shall be as shown on figure 1.

3.3.3.2 *Shearing section.* The drive shall incorporate a shearing section to prevent damage to the engine in case of pump seizure. Failure of the shearing section shall not result in damage to any other part of the drive, pump, or engine. It shall be possible to replace a failed shearing section without machining or hand-fitting. The shearing section shall fail under any applied torque greater than the applicable value listed in table I.

3.3.4 *Shaft seal.* A positive shaft seal shall be incorporated within the pump to limit the leakage of air out of the pump and leakage of oil out of or into the pump. The leakage of air out of the pump seal shall be no more than 0.05 cubic foot per minute (cfm) under any possible combination of speeds and air pressure differentials within the requirements of this specification. The leakage of oil into or out of the pump shall not exceed 10 milliliters per hour under any possible combination of speeds and air pressure differentials within the requirements of this specification, when an oil pressure from 0 to 1 pound per square inch gage (psig) exists in the chamber between the pump shaft seal and the engine. If venting provisions are necessary to effect the seal, they shall be built into the pump and shall be automatically effective for either direction of rotation.

3.3.5 *Oil dilution connection.* A ⅛-inch pipe tap opening shall be provided on each port for admitting fuel to dilute the oil in the pump. These openings shall be located as shown on the applicable detail AN standard.

3.3.6 *Lubrication.* Lubricating oil shall conform to grade 1100 of Specification MIL–L–6082. A means of metering the oil, which will be supplied at a pressure of 35 to 100 pounds per square inch (psi), from the engine pressure oil supply, shall be built into the pump. The device shall meter sufficient oil to assure lubrication throughout the operating range of the pump, but the oil metered shall not exceed the maximum specified in table I at 50 psi, nor be less than the minimum specified by the manufacturer at 35 psi. Provision shall be made for supplying oil to the metering device from the pressure oil hole located in the mounting pad, when the pump is installed in any position. An external oil connection to the metering device consisting of a ⅛-inch pipe thread tapped boss shall also be provided.

3.3.7 *Balance.* Vibration of pump shall not exceed 2.5 mils (.0025-inch) under any conditions of operation at speeds up to and including 120 percent of rated speed. To insure this condition of operation, at least two through vanes or four single vanes in vane type pumps shall be used.

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TABLE I. *Pump characteristics.*

Part No.	Minimum capacity cfm of pump inlet air at inlet pressure 23.5 inches mercury absolute (Hg.abs.)		Maximum driving torque (pounds-inch) at inlet pressure 22.5 inches Hg.abs.	
	Outlet pressure 31.5 inches Hg.abs. 1,500 rpm, ± 15 rpm	Outlet pressure 46.5 inches Hg.abs. 2,250 rpm, ± 22 rpm	Outlet pressure 35.5 inches Hg.abs.	Outlet pressure 49.5 inches Hg.abs.
AN6110–1	5.5	.....	20	.....
AN6111–1	9.0	12.0	.....	40
AN6111–2	9.0	12.0	.....	40
AN6112–1	.....	18	.....	80

Part No.	Shearing section torque	Max. oil flow at 50 psi	Weight	Mounting pad and drive	Gasket
	Pounds-inch (max.)	Cubic centimeters per hour	Pounds (max.)	AND No.	Part No.
AN6110–1	800	100	4.25	AND10000	AN4045–1
AN6111–1	800	150	5.80	AND10000	AN4045–1
AN6111–2	800	150	6.30	AND10001	AN4044–1
AN6112–1	1,500	180	9.90	AND10001	AN4044–1

3.3.8 *Internal passages.* The porting arrangement within the pump shall be such that a quantity of oil sufficient to cause failure of the pump by hydraulic lock cannot be trapped therein under any conditions of operation.

3.3.9 *Threaded parts.* Screw threads shall be in accordance with class 3 fit of Specification MIL–7742. Pipe threads shall be in accordance with Specification MIL–P–7105. All internal and external threaded parts shall be positively locked or safetied.

3.3.10 *Screen.* A #10 mesh screen of #24 Roebling steel wire gage wire, or equal, shall be included to protect the moving parts for either direction of pump rotation, from foreign matter entrained in the air.

3.3.11 *Mounting flange, drive, and wrench clearance.* The mounting flange, drive, and wrench clearance shall be in accordance with figure 1, 2, 3, or 4, as applicable.

3.3.12 *Gasket.* Pump shall be provided with a mounting flange gasket as specified in table I.

3.3.13 *Interchangeability of parts.* All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL–D–70327, or as otherwise specified by the procuring activity.

3.3.14 *Standard parts.* AN or MS standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawings by their part numbers. Commercial utility parts such as screws, bolts, nuts, and cotter pins may be used provided they have suitable properties and are replaceable by standard parts (AN or MS) without alteration. If no suitable standard part is in effect on date of invitation for bids, commercial parts may be used provided they conform to this specification.

3.3.15 *Metals.* Where practicable, all metals used in the construction of vacuum pump shall be corrosion-resistant or shall be suitably protected to resist corrosion during normal service life. The use of dissimilar metals, especially brass, copper, or steel in intimate metal-to-metal contact with alu-

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minum or aluminum alloy, shall be avoided wherever practicable. Magnesium and magnesium alloys shall not be used.

3.4 Size and weight. Pump dimensions shall conform to the applicable detail AN standard (see 1.2). Weight shall not exceed the value specified in table I.

3.5 Performance.

3.5.1 *Capacity.* The minimum capacities prior to the endurance test (see 4.7.8) shall be as specified in table I. The allowable variation in capacity shall be ± 5 percent of the average capacity, which shall be the average of the capacities of the first 10 production pumps tested, that have capacities varying not more than 3 percent. These 10 pumps shall be calibrated for capacity with air inlet temperature varying from 16° C. (60° F.) to 38° C. (100° F.), and an average curve plotted. Curves shall also be plotted on the same sheet, representing the upper and lower capacity limits of ± 5 percent of the average curve. For pumps having two capacity ratings, the tolerance on the higher capacity rating shall apply and the lower capacity rating shall be used only as a minimum. The pump shall have the required capacity in either direction of rotation.

3.5.2 *Temperature rise and oil flow.* The maximum oil flow shall not exceed the value specified in table I and the minimum shall not be less than that of the sample pumps subjected to preproduction test (see 4.4.1), as determined at 50 psi. The allowable variation of temperature rise prior to the endurance test shall be within ± 6° C. (10° F.) of the average temperature rise, which shall be the average of the temperature rises of the first 10 production pumps tested that have temperature rises varying not more than 5° F. These 10 pumps shall be calibrated for temperature rise with the air inlet temperature varying from 16°C. (60° F.) to 38° C. (100° F.), and an average curve plotted for submission to the procuring activity. Curves shall also be plotted on the same sheet, representing the upper and lower temperature rise limits of ± 6° C. (± 10° F.) of the average curve.

3.5.3 *Air and oil leakage.*

3.5.3.1 *Pump.* The pump shall not leak oil under any condition of operation up to 100 psi oil pressure, except from the oil seal.

3.5.3.2 *Shaft seal.* The air leakage out of the shaft seal shall not exceed 0.05–cfm and the oil leakage into or out of the shaft seal shall not exceed 10 milliliters per hour at any time during the test. The total metered oil and shaft seal leakage oil shall not exceed the maximum oil flow allowed.

3.5.4 *Torque.* The torque required to drive the vacuum pump at any speed up to and including 120 percent rated speed shall not exceed the value specified in table I.

3.5.5 *Low-temperature operation.* The vacuum pump shall operate in a satisfactory manner during and following the low-temperature operation test (see 4.7.5).

3.5.6 *Shearing section.* The torque required for failure of the shearing test section shall not exceed the value specified in table I.

3.5.7 *Calibration.* The pump shall operate satisfactorily when subjected to all tests specified in section 4. Information obtained in the calibration test shall be recorded in graphical form.

3.5.8 *Endurance.* The vacuum pump shall satisfactorily withstand the endurance test (see 4.7.8), with no failure of parts and no evidence of excessive wear.

3.5.9 *Recalibration after endurance test.* The capacity of the pump shall not have decreased by more than 10 percent after the endurance test. The temperature rise shall not have increased more than 6° C. (10° F.) and the oil flow shall not change more than ± 25 percent, nor exceed the maximum allowable limit. The shaft seal leakage shall not exceed 10 cubic centimeters per hour. There shall be no other oil leakage from the pump.

3.6 Finish.

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3.6.1 *Plating.* Steel parts in contact with aluminum or aluminum alloy shall be cadmium-plated in accordance with Specification QQ–P–416, or zinc-plated in accordance with Specification QQ–Z–325.

3.6.2 *Anodizing.* All aluminum alloy parts shall be anodized where applicable in accordance with Specification MIL–A–8625.

3.6.3 *Paint finish.* All external surfaces, except corrosion-resistant steel, anodized aluminum alloy, working surfaces, threads, and cadmium-plated parts, shall be finished with one coat of primer conforming to Specification MIL–P–8585 and two coats of black enamel conforming to Specification MIL–E–5557.

3.7 **Identification of pump.** Pump shall be identified in accordance with Standard MIL–STD–130.

3.8 **Workmanship.** The vacuum pump assemblies, including all detail parts, shall be fabricated and finished to produce an item free from defects which could affect proper functioning of the complete unit in service.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Supplier.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Inspection.** Each pump shall be examined to determine conformance with the applicable standards (see 2.1), and with this specification with respect to material, workmanship, design, and construction.

4.3 **Sampling.**

4.3.1 *Lots.* A lot shall consist of all pumps manufactured under the same conditions and submitted for acceptance at one time.

4.3.2 *Routine tests.* Samples for routine representative tests shall consist of one vacuum pump selected from the first 10, one from the next 90, and one from the next 900, and one from every 5,000, after the first 1,000 pumps have been accepted. If a pump selected for routine tests fails to conform to this specification, 10 other pumps shall be selected at random from the lot represented and submitted to all the tests specified. If any pump of the group of 10 fails to meet these tests, the entire lot shall be rejected.

4.4 **Classification of tests.** Inspection and tests of pumps shall be classified as preproduction and acceptance.

4.4.1 *Preproduction.* Preproduction samples shall be subjected to all the tests specified in this specification. Sample vacuum pumps shall have the oil metering device adjusted so that oil flow is the minimum which the manufacturer desires. Upon completion of the preproduction testing, a test report prepared in accordance with Specification MIL–T–9107, shall be forwarded to the contracting officer, and approval obtained before proceeding with the manufacture of pumps on contract or order.

4.4.2 *Acceptance.* Acceptance tests are those tests performed on individual pumps to determine conformance with this specification. These tests shall consist of the tests in 4.4.2.1 and 4.4.2.2.

4.4.2.1 *Individual.* Each pump shall be subjected to the following tests. Failure of any pump to pass these tests shall be cause for rejection of the individual pump.

(a) Inspection (see 4.2).
(b) Capacity (see 4.7.1).
(c) Temperature rise and oil flow (see 4.7.2).
(d) Air and oil leakage (see 4.7.3).

4.4.2.2 *Selected samples (see 4.3).* In addition to the tests in 4.4.2.1, each routine representative sample shall be subjected to the following tests, in the order given:

(a) Torque (see 4.7.4).
(b) Low-temperature operation (see 4.7.5).

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(c) Endurance (see 4.7.8).
(d) Recalibration (see 4.8).
(e) Shearing section (see 4.7.6).

4.5 Test reports.

4.5.1 *Preproduction.* Reports for the preproduction tests shall include, in addition to the information required in test reports for routine representative samples (see 4.7.7), graphically recorded, showing all pertinent data.

4.5.2 *Acceptance.* A record showing quantitative results for all acceptance tests shall be maintained.

4.5.3 *Routine.* A record showing quantitative results for all routine tests of representative samples (see 4.3), shall be maintained. These test reports shall include the following:

(a) Results of capacity and temperature rise and oil flow, compared with approved average (see 4.7.1 and 4.7.2).
(b) Average of pump temperature, pump air inlet temperature, and pump speed during low-temperature test (see 4.7.5).
(c) Tabulation of all readings taken during endurance test (at least two readings in each 24-hour period), and of averages for each speed (see 4.7.8).
(d) Recalibration test results compared with the original capacity, temperature rise and oil flow, and air and oil leakage tests of the same pump.
(e) Results of shearing section test (see 4.7.6).

4.6 Preparation for tests.

4.6.1 *Cleaning.* All oil and grease or other corrosion-resistant compounds shall be removed from interior and exterior parts of the pump.

4.6.2 *Direction of rotation.* Unless otherwise specified, each pump shall be assembled and tested with clockwise rotation, when viewed from the anti-drive end.

4.6.3 *Oil.* Unless otherwise specified, all tests shall be conducted using lubricating oil conforming to grade 1100 of Specification MIL–L–6082.

4.6.4 *Definitions and measurements.*

4.6.4.1 *Inlet pressure (P_1).* The pump inlet pressure shall be taken at the point located as shown on figure 5.

4.6.4.2 *Outlet pressure (P_2).* The pump outlet pressure shall be taken as shown on figure 5.

4.6.4.3 *Inlet temperature (T_1).* The pump inlet temperature shall be taken as shown on figure 5.

4.6.4.4 *Outlet temperature (T_2).* The outlet temperature shall be taken as shown on figure 5.

4.6.4.5 *Pump inlet air.* Pump inlet air is defined as the air at the inlet of the pump. The density of the air, ρ_1 shall be based on the air pressure P_1 and air temperature T_1.

4.6.4.6 *Pump capacity.* Pump capacity in terms of "pump inlet air" shall be the weight flow as measured by the flow meter divided by ρ_1.

4.6.4.7 *Free air.* Free air shall be considered as air existing at the conditions of temperature and barometeric pressure in the room in which the test is conducted.

4.6.5 *Test set-up.* The test set-up for all tests except the low temperature test, shall conform to figure 5. Unless otherwise specified, the oil pressure supplied to the metering device shall be 50 psi.

4.6.6 *Break-in run.* Prior to conducting the first test (capacity), the pump shall be subjected to a break-in run consisting of 2 hours of running with as little restriction in the pump inlet and outlet lines as possible. This test shall be conducted as follows:

One hour at 2,250 to 2,750 rpm.
One hour at 3,500 to 4,000 rpm.

A total run-in up to a maximum of 5 hours

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may be conducted on the pump at the discretion of the manufacturer.

4.7 **Test methods.** The shaft oil seal leakage and other oil leakage shall be observed during all tests. Unless otherwise specified, the following tests shall be conducted in the order given:

4.7.1 *Capacity.* Each pump shall be operated at the conditions specified under minimum capacity in table I. The capacity of the pump being tested shall be checked at the exact inlet temperature involved during the test. The capacity shall be measured after 5 minutes of operation, or at such time as the outlet temperature reaches its terminal temperature, whichever is the greater. The capacity shall be determined for each direction of rotation. The capacity shall be corrected to the speed specified, assuming capacity proportional to speed.

4.7.1.1 For pumps having two capacity ratings, the entire procedure shall be carried out for each rating.

4.7.2 *Temperature rise and oil flow.* Each pump shall be run at 3,750 rpm, ± 50 rpm, oil pressure 50 psi, and the inlet and outlet pressures specified in table I for driving torque. This run shall continue for 5 minutes, or longer if necessary for the outlet temperature to reach its terminal value. The temperature rise of the pump being tested shall be checked at the exact inlet temperature involved during the test. The inlet and outlet temperatures shall then be noted and the temperature rise and oil flow determined.

4.7.3 *Air and oil leakage.*

4.7.3.1 *Pump.* The pump shall be run at the conditions specified for temperature rise and oil flow, except that an oil pressure of 100 psi shall be maintained. The oil seal leakage shall be determined and the pump shall be examined for leakage.

4.7.3.2 *Oil seal.* The pump shall be run at 3,000 rpm, 22½ inches mercury inlet pressure absolute, and a maximum of 1-inch mercury outlet pressure gage, with the chamber between the pump seal and laboratory test drive filled with oil at a pressure of 1 psig. The oil leakage into the pump through the oil seal shall be determined.

4.7.3.3 *Oil leakage, pump idle.* An oil pressure of 1 psig between the pump seal and the laboratory test drive shall be maintained when the pump is idle with atmospheric pressure at the inlet and outlet ports. The oil leakage into the pump through the oil seal shall be determined.

4.7.3.4 *Air seal.* The pump shall be run at 3,000 rpm, 28½ inches mercury absolute minimum and discharge pressure 49½ inches mercury absolute. The air leakage through the pump seal into the laboratory test drive shall be determined.

4.7.4 *Torque.* The torque required to drive the pump under the conditions specified in table I shall be measured.

4.7.5 *Low-temperature operation.* The pump shall be run at room temperature until it is well permeated with oil. The intake and outlet ports shall be connected, and the pump maintained at a temperature from —51° C. (—60° F.) to —57° C. (—70° F.) throughout the remainder of this test. After a minimum of 2 hours from the time the pump has reached —60° F., the pump shall be started, brought up to 750 rpm, ± 50 rpm, and allowed to run approximately 30 seconds. This shall be repeated for a total of 50 starts, allowing at least 5 minutes between starts, and turning the pump over twice by hand before each start. The starts shall be made with as low an inlet suction and outlet pressure as practicable. The oil pressure shall be approximately 35 psi during the last 15 seconds of each start. During the entire test the pump shall be lubricated with a mixture of 30-percent gasoline and 70-percent oil. Gasoline shall conform to grade 130 of Specification MIL–G–5572, and lubricating oil to grade 1100 of Specification MIL–L–6082.

4.7.6 *Shearing section.* The pump drive or a shearing section test piece shall be tested to determine the torque required for failure of the shearing section. Shearing section

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test pieces shall not be used for this test on routine representative samples; the pump drive shall be subjected to the test.

4.7.7 *Calibration.* The capacity, torque, inlet temperature, and outlet temperature shall be determined at a shaft speed of 1,500 rpm at each of the inlet pressures and pressure differentials specified in table II as limited by the maximum permissible temperature rise at high suction. The tests shall be repeated for shaft speeds of 2,250, 3,000, 3,750, and 4,500 rpm.

TABLE II. *Pressures for calibration test.*

AN Part No.	Inlet pressure in. hg. abs.	Outlet pressure in. hg. greater than inlet pressure
AN6110–1	18,22,26	6,12,18
All other pumps	18,22,26	6,16,26

4.7.8 *Endurance.* The oil pressure during the endurance test, with the exception of the 2-hour oil film maintenance test, shall be maintained at 35 psi. During the entire endurance test, the pump shall be installed on the laboratory mounting pad and a shim placed between one edge of the pump mounting flange and the face of the mounting pad so that the included angle will be ¼-degree. Readings of speed, inlet and outlet pressure, inlet and outlet temperature oil flow, oil pressure, oil seal leakage, and barometric pressure shall be recorded at least twice in each 24-hour period throughout the endurance test.

4.7.8.1 *Universal joint drive.* The pump shall be driven by a variable velocity universal joint drive. The universal joint drive may include either one or two universal joints which shall be of a type having the axes of the trunnion bearings in a common plane. The angles, as listed in table III, apply to speeds as shown for both single and double universal joint drives. Single joint drives shall be set up with the joint shafts at the angle shown. Double joint drives shall be set up with both joint shafts at the angle shown and with the driving and driven shafts of the entire universal joint drive, parallel.

4.7.8.2 *Inertia of drive side of universal joint drive.* The total moment of inertia of the drive side of the universal joint drive shall be at least 20 times that of the pump rotors, and the parts of the universal joints subjected to variable velocity, or sufficiently great so that the torsional oscillations of the drive side shall be less than 10 percent of the torsional oscillations of the driven shaft side. The entire universal joint drive between the drive side and the pump, including the connection to the pump, shall be torsionally rigid.

4.7.8.3 *Conditions and duration.* The pump shall be subjected to the following conditions of operation, for the periods specified:

(a) Fifty hours of operation at 23.5 inches hg. abs. inlet pressure and 31.5 inches hg. abs. outlet pressure at a pump speed from 1,500 to 1,690 rpm.

(b) Fifty hours of operation at 23.5 inches hg. abs. inlet pressure, speed 2,250 to 2,440 rpm, and outlet pressure as follows:

For AN6110–1 pumps: 34 inches hg. abs.

For all other pumps: 48.5 inches hg. abs.

(c) Four hundred hours at 22.5 inches hg. abs. inlet pressure, speed 3,750 to 3,940 rpm, and outlet pressure as follows:

For AN6110–1 pumps: 35.5 inches hg. abs.

For all other pumps: 49.5 inches hg. abs.

(d) Fifty hours at 22.5 inches hg. abs. inlet pressure, speed 4,125 to 4,210 rpm, and outlet pressure as follows:

For AN6110–1 pumps: 35.5 inches hg. abs.

For all other pumps: 49.5 inches hg. abs.

4.7.8.4 *Oil film maintenance.* The pump shall then be run for 15 minutes with oil conforming to grade 1100 of Specification MIL–L–6082, supplied to the oil metering device at a pressure of 50 psi. Following

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TABLE III. *Angles for universal joint drive.*

Pump speed in rpm	Joint angle degrees (tolerance, + 1½ — 0)		Torsional vibration amplitude (± degrees)
	Single joint drive	Double joint drive	
500 to 3,000	20	15	2
3,000 to 4,314	15	10	1

this, the pump shall be run for 2 hours at 3,000 rpm, 4.0 inches hg. suction and 2.0 inches hg. pressure, without adding any oil through the oil metering device or by any other means.

4.7.8.5 *Determination of wear.* After the tests specified in 4.7.8.3 and 4.7.8.4, all parts shall be inspected and measured to determine wear.

4.8 **Recalibration.** Following the endurance test, the air pump shall be recalibrated in accordance with the method specified herein under *capacity, temperature rise and oil* flow, and oil leakage.

4.9 **Inspection of preparation for delivery.** Preservation, packaging, packing, and marking shall be inspected to assure compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 **Preservation and packaging.** Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level* A. Each pump shall be preserved in accordance with method IA–8 of Specification MIL–P–116. Six cubic centimeters of engine preservative oil conforming to grade 2 of Specification MIL–L–21260 shall be poured into one of the ports. The coupling shall be turned at least six times to insure complete lubrication, and the pump then drained. All openings shall be capped or plugged with closures conforming to Specification MIL–C–5501, to prevent oil leakage. The protruding coupling shaft of each pump shall be coated with preservative material conforming to grade 2 of Specification MIL–C–16173 or to class 3 of Specification MIL–C–11796. Pumps shall be individually packaged in unit containers conforming to Specification PPP–B–636.

5.1.2 *Level C.* Each pump shall be processed as follows: Six cubic centimeters of engine preservative oil conforming to Specification MIL–L–21260 shall be poured into one of the ports. The coupling shall be turned at least six times to insure complete lubrication, and the pump then drained. All openings shall be capped or plugged with closures conforming to Specification MIL–C–5501, to prevent oil leakage. Pumps shall then be preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration, and damage during handling and shipment from supply source to the first receiving activity.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Pumps packaged as specified in 5.1 shall be packed in snug-fitting cleated-plywood boxes conforming to Specification PPP–B–601, oversea type, using type II, class 2 plywood, or in nailed wood boxes conforming to class 2 of Specification PPP–B–621 for type 2 load, or in fiberboard boxes conforming to class 2 of Specification PPP–B–636. Boxes shall be closed and strapped or banded in accordance with the applicable box specification, or appendix thereto; straps, when specified, shall conform to type I, class B of Specification QQ–S–781.

5.2.2 *Level B.* Pumps packaged as specified in 5.1 shall be packed in snug-fitting wirebound wood boxes conforming to Specification PPP–B–585 for class 2 use, or in cleated-plywood boxes conforming to Specification PPP–B–601, oversea type, using type

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III, class 1 plywood, or in nailed wood boxes conforming to class 1 of Specification PPP–B–621 for type 2 load, or in fiberboard boxes conforming to class 1 of Specification PPP–B–636. Boxes shall be closed and strapped or banded in accordance with the applicable box specification, or appendix thereto; straps, when specified, shall conform to type I, class B of Specification QQ–S–781.

5.2.3 *Level C.* Pumps packaged as specified in 5.1 shall be placed in shipping containers prepared for delivery in a manner to insure carrier acceptance and safe delivery at the destination. Containers shall be in accordance with Uniform Freight Classification Rules or with regulations of other carriers applicable to the mode of transportation.

5.3 **Marking.** In addition to any special marking required by the contract or order, marking shall be in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.** Power driven rotary vacuum pump is intended for use on aircraft to provide vacuum for instruments, and pressure for operation of de-icer systems.

6.2 **Ordering data.** Procurement documents should specify the following:

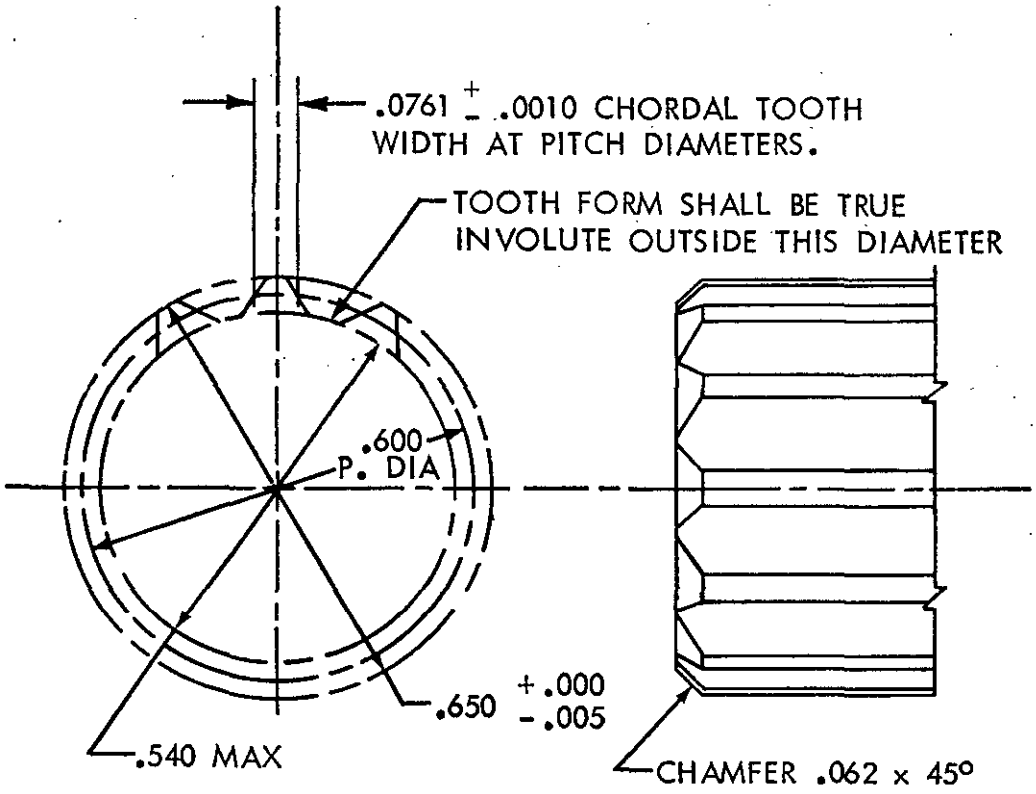
(a) Title, number, and date of this specification.
(b) AN part number required (see 1.2).
(c) Level of preservation and packaging, and of packing required (see 5.1 and 5.2).
(d) Whether special marking is required (see 5.3).

Custodians:
Army—MO
Navy—Weps
Air Force—OCA
User activities:
Army—MO
Navy—Weps
Air Force—OCA

Preparing activity:
Army—MO(AVSCOM)
Project No. 4320–0041

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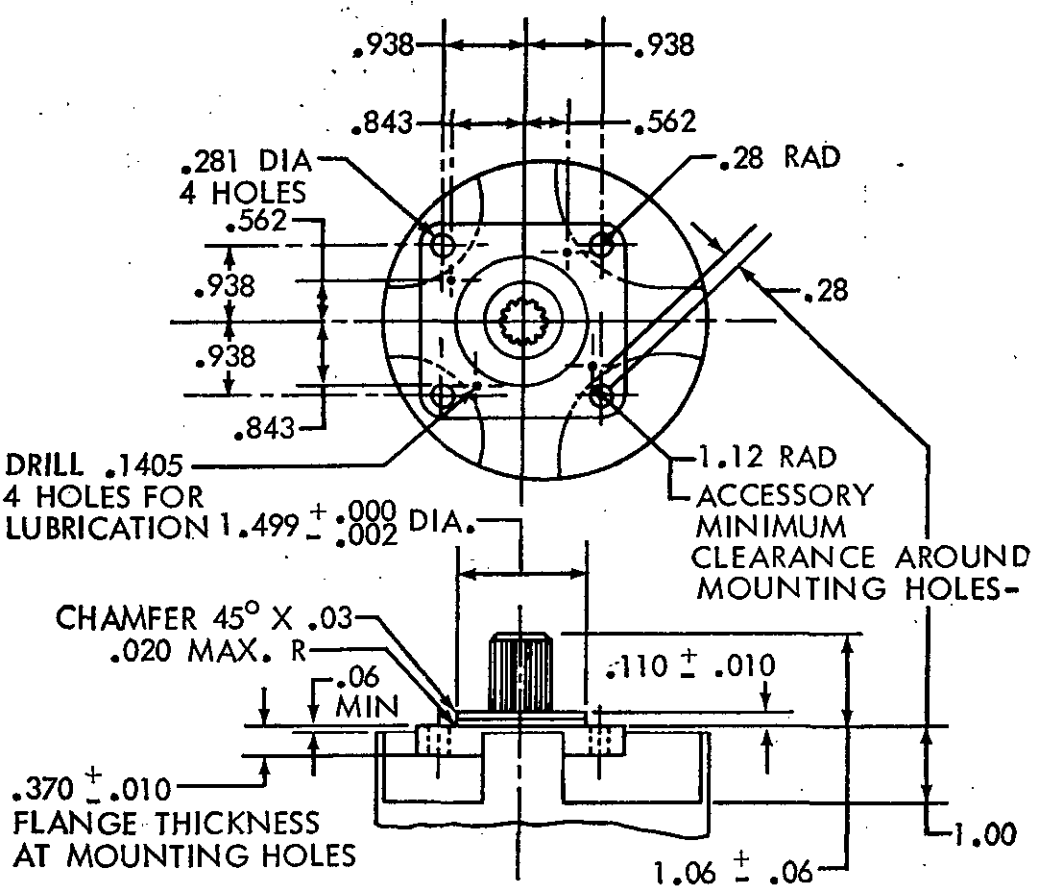


SPLINE DATA: 12 TEETH, 20/40 PITCH, 30° PRESS. ANGLE.

FIGURE 1. *Spline details—pumps.*

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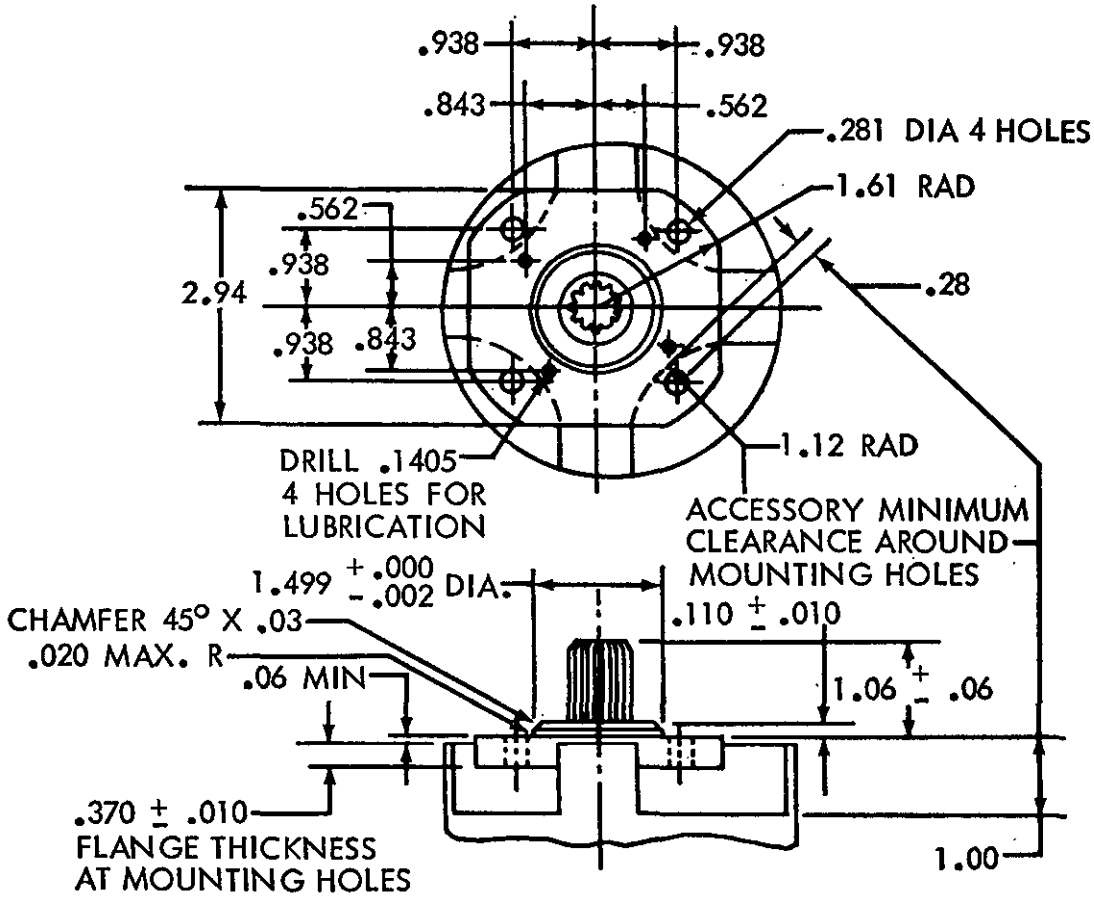
REMOVE BURRS AND SHARP EDGES.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: TWO PLACE DECIMALS ± .02, THREE PLACE DECIMALS ± .005.

FIGURE 2. *Mounting flange, drive, and wrench clearance for Pump AN6110-1.*

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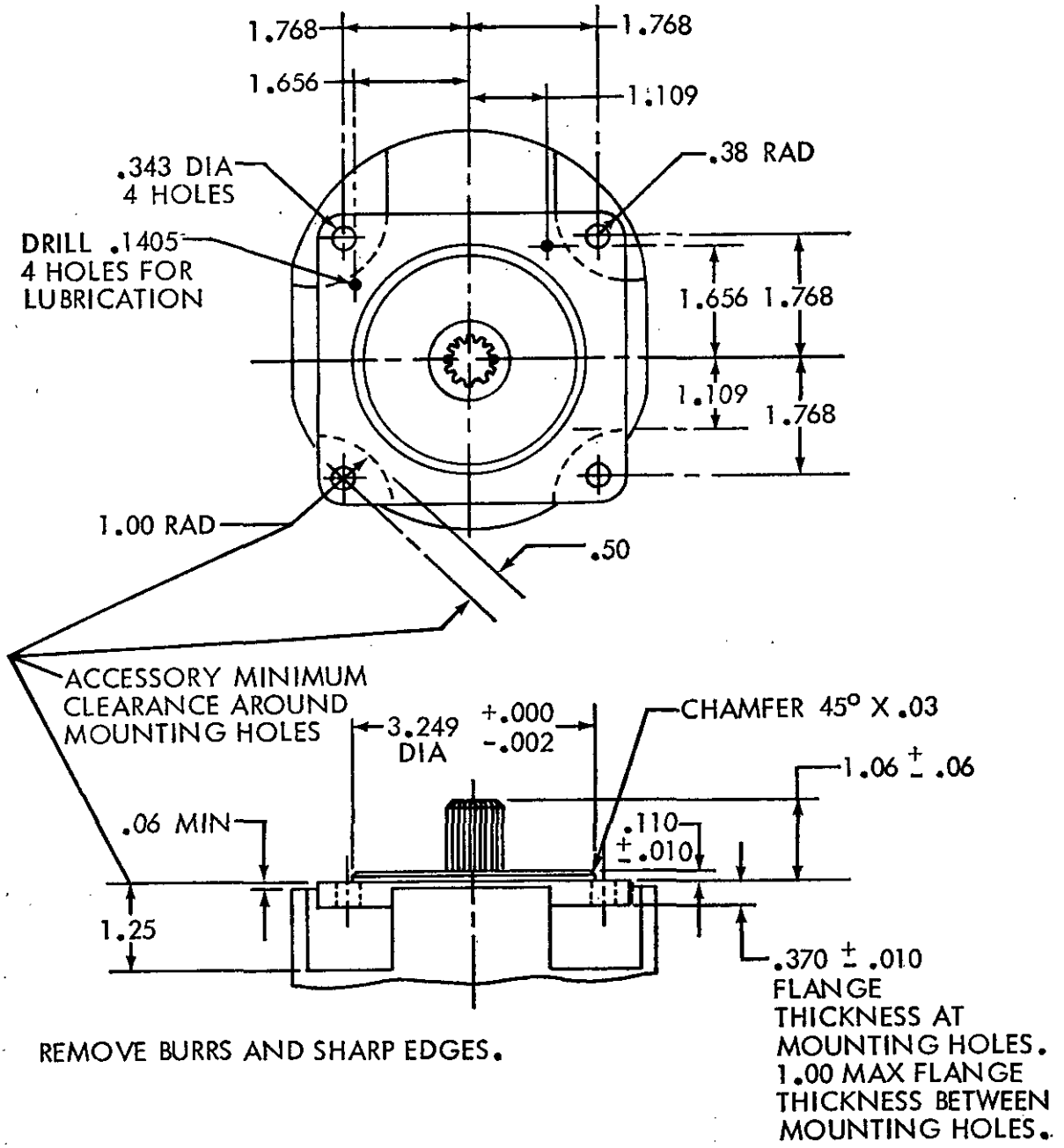
REMOVE BURRS AND SHARP EDGES.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: TWO PLACE DECIMALS ± .02, THREE PLACE DECIMALS ± .005.

FIGURE 3. *Mounting flange, drive, and wrench clearance for Pump AN6111-1.*

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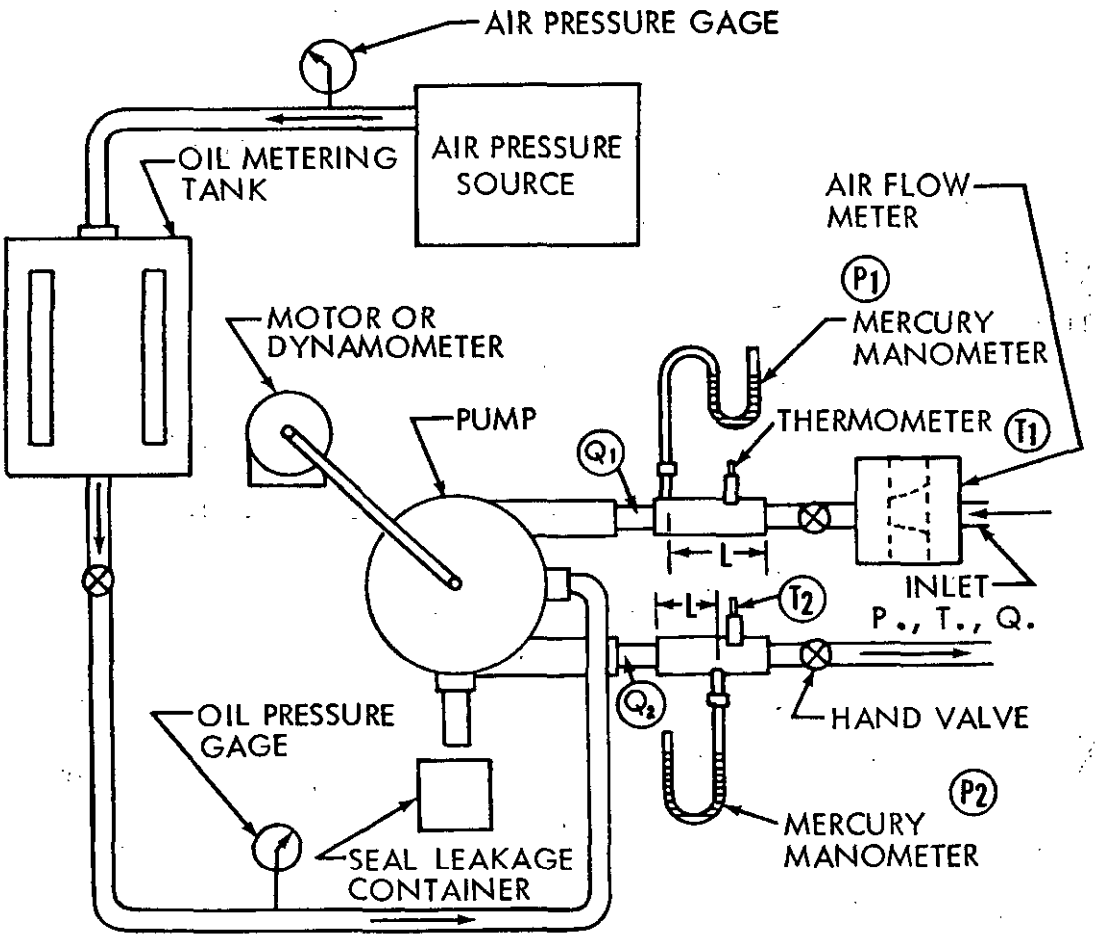


DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES:
2 PLACE DECIMALS ± .02, 3 PLACE DECIMALS ± .005.

FIGURE 4. *Mounting flange, drive, and wrench clearance at mounting holes for Pump AN6111-2 and AN6112-1.*

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POINT OF MANOMETER CONNECTION SHALL BE PRECEDED BY A STRAIGHT SECTION OF FITTING OF MINIMUM LENGTH "L" AND "I.D." AS SHOWN BELOW:

DIA. OF PUMP PORT OPENING (INCHES)	INSIDE DIAMETER OF FITTING (INCHES)	MINIMUM LENGTH OF FITTING (INCHES) "L"
3/8	.42	2-1/2
1/2	.54	3
5/8	.65	4
3/4	.90	5-1/2
1	1.15	7

SYMBOLS P, T AND Q ARE PRESSURE IN P.S.I., TEMPERATURE IN°F., AND AIR FLOW IN C.F.M., RESPECTIVELY. SUBSCRIPTS O, 1, AND 2, REFER TO FREE AIR, INLET, AND OUTLET CONDITIONS, RESPECTIVELY

FIGURE 5. *Pump test set up.*

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